NexusMart Data Warehouse: Star Schema

PostgreSQL schema mart | 1 fact table + 5 dimensions | CODE//CONNECT'26 Data Analytics

mart.dim_date — Date dimension

PK	**date_key**	int
	full_date	date
	year_num	int
	quarter_num	int
	month_num	int
	month_name	text
	week_of_year	int
	day_name	text
	is_weekend	bool
	is_ramadan	bool
	is_eid_ul_fitr	bool
	is_eid_ul_adha	bool
	is_14_august	bool
	is_11_11	bool
	is_black_friday	bool
	is_order_period	bool

mart.dim_customer — Customer dimension (SCD Type 1)

PK	**customer_key**	int
	customer_id	text
	full_name	text
	email	text
	phone	text
	gender	text
	registration_date	date
	birth_year	int
	segment	text
	city	text
	province	text
	region	text

mart.dim_product — Product dimension

PK	**product_key**	int
	product_id	text
	product_name	text
	category	text
	sub_category	text
	brand	text
	list_price	numeric
	weight_kg	numeric

mart.dim_seller — Seller dimension

PK	**seller_key**	int
	seller_id	text
	seller_name	text
	join_date	date
	seller_tier	text
	city	text
	province	text
	rating	numeric

mart.fact_sales — Fact table | grain: one row per order line item

PK	**order_item_id**	text
	order_id	text
FK	**date_key**	int
FK	**customer_key**	int
FK	**product_key**	int
FK	**seller_key**	int
FK	**location_key**	int
	quantity	int
	unit_price	numeric
	gross_amount	numeric
	discount_amount	numeric
	net_amount	numeric
	cost_amount	numeric
	shipping_fee	numeric
	shipping_days	int
	is_cancelled	bool
	is_returned	bool
	returned_amount	numeric
	return_date_key	int
	return_reason	text
	return_status	text
	order_status	text
	payment_method	text
	channel	text

mart.dim_location — Location dimension

PK	**location_key**	int
	city_id	text
	city	text
	province	text
	region	text
	population_tier	text

1 — ***** : dim_date.date_key → fact_sales.date_key

1 — ***** : dim_customer.customer_key → fact_sales.customer_key

1 — ***** : dim_product.product_key → fact_sales.product_key

1 — ***** : dim_seller.seller_key → fact_sales.seller_key

1 — ***** : dim_location.location_key → fact_sales.location_key

Design notes

1. Fact grain: one row per order line item. Amounts in PKR.
2. Every dimension has an Unknown member (key = -1); orphan keys and blank ship city map there.
3. Dimensions are not linked to each other; city/province/region are copied in.
4. Order shipping_fee is split across its lines by gross (no double count).
5. Valid (non-Rejected) returns are aggregated per line before joining.
6. Truncate-and-load (idempotent): clean -> dim_* -> fact_sales -> checks.

Legend

PK — primary key (surrogate key in dimensions)

FK — foreign key to a dimension

1 ---- * one dimension row : many fact rows